

### 3. Force and Pressure

#### 1. Write proper word In the blank space:

**(a) The SI unit of force is the .....**

(dyne, newton, joule)

Answer:

The SI unit of force is the newton.

**(b) The air pressure on our body is equal to the ..... pressure.**

(atmospheric, sea bottom, space)

Answer:

The air pressure on our body is equal to the atmospheric pressure.

**(c) For a given object, the buoyant force in liquids of different ..... is ..... .**

(the same, density, different, area)

Answer:

For a given object, the buoyant force in liquids of different density is the same.

**(d) The SI unit of pressure is .....**

( $\text{N/m}^3$ ,  $\text{N/m}^2$ ,  $\text{kg/m}^2$ ,  $\text{Pa/m}^2$ )

Answer:

The SI unit of pressure is  $\text{N/m}^2$ .

## 2. Make a match.

| Group 'A'           | Group 'B'                          |
|---------------------|------------------------------------|
| 1. Fluid            | a. Higher pressure                 |
| 2. Blunt knife      | b. Atmospheric pressure            |
| 3. Sharp needle     | c. Specific gravity                |
| 4. Relative density | d. Lower pressure                  |
| 5. Hectopascal      | e. Same pressure in all directions |

Answer:

| Group 'A'           | Group 'B'                          |
|---------------------|------------------------------------|
| 1. Fluid            | e. Same pressure in all directions |
| 2. Blunt knife      | d. Lower pressure                  |
| 3. Sharp needle     | a. Higher pressure                 |
| 4. Relative density | c. Specific gravity                |
| 5. Hectopascal      | b. Atmospheric pressure            |

## 3. Answer the following questions in brief.

**(a) A plastic cube is released in water. Will it sink or come to the surface of water?**

Answer:

It will come to the surface of water.

[Note: This is because its density is less than that of water. When it floats, the unbalanced force acting on it is zero.]

**(b) Why do the load carrying heavy vehicles have large number of wheels?**

Answer:

The pressure produced by a given force depends on the area of the surface on which the force acts. The greater the surface area, the less is the pressure produced. Load carrying heavy vehicles have large number of wheels so that the load (weight, force) is distributed over large surface area of the wheels in contact with the road. Hence, the pressure decreases and the tyres do not burst.

**(c) How much pressure do we carry on our heads? Why don't we feel it?**

Answer:

The air pressure at the sea level is about  $101 \times 10^3$  Pa. This is the pressure that we carry on our heads. The cavities in our body are filled with air, and arteries and veins are filled with blood. Their pressure balances the pressure due to the atmosphere. Hence, we don't feel the atmospheric pressure.

#### **4. Why does it happen?**

**(a) Why does it happen? A ship dips to a larger depth in freshwater as compared to marine water.**

Answer:

The buoyant force acting on a body is proportional to the density of the fluid in which the body is immersed. The density of freshwater is less than that of marine water. Hence, the buoyant force on a body in freshwater is less than that in marine water. Therefore, a ship dips to a larger depth in freshwater as compared to marine water.

**(b) Why does it happen? Fruits can easily be cut with a sharp knife.**

Answer:

1. It is easy to cut vegetables. fruits with a sharp knife. A blunt knife does not work here.
2. The force exerted perpendicularly on a unit area is called 'pressure'

$$\text{Pressure} = \frac{\text{Force}}{\text{Area on which the force is applied}}$$

3. Presently we are considering only the force acting on an area in a direction perpendicular to it.

**(c) Why does it happen? The wall of a dam is broad at its base.**

Answer:

1. The pressure at a point in a liquid is proportional to the height of the liquid column above it. Hence, the pressure of water in a dam is much greater at the bottom of the dam than at the top.
2. To withstand this high pressure, the wall of a dam is made stronger and thicker (broad) at the base than at the top.

**(d) Why does it happen? If a stationary bus suddenly speeds up, passengers are thrown in the backward direction.**

Answer:

1. When passengers sit or stand in a stationary bus, they are in a state of rest. When the bus suddenly speeds up, the lower (parts of their body in contact with the bus acquire the speed of the bus.
2. The upper parts of their body, however, continue to be in the state of rest due to inertia. Hence, they are thrown in the backward direction.

5. Complete the following tables.

| Mass (kg) | Volume (m <sup>3</sup> ) | Density (kg/m <sup>3</sup> ) |
|-----------|--------------------------|------------------------------|
| 350       | 175                      | -                            |
| -         | 190                      | 4                            |

Answer:

Using the formula,

density = mass/volume:

| Mass (Kg) | Volume (m <sup>3</sup> ) | Density (Kg/m <sup>3</sup> ) |
|-----------|--------------------------|------------------------------|
| 350       | 175                      | 2                            |
| 760       | 190                      | 4                            |

| Density of Metal (kg/m <sup>3</sup> ) | Density of water (kg/m <sup>3</sup> ) | Relative Density |
|---------------------------------------|---------------------------------------|------------------|
|                                       | 10 <sup>3</sup>                       | 5                |
| 8.5 x 10 <sup>3</sup>                 | 10 <sup>3</sup>                       | -                |

Answer:

Using the formula, relative density = density of a metal/density of water:

| Density of Metal (Kg/m <sup>3</sup> ) | Density of water (Kg/m <sup>3</sup> ) | Relative Density |
|---------------------------------------|---------------------------------------|------------------|
| 5 x 10 <sup>3</sup>                   | 10 <sup>3</sup>                       | 5                |
| 8.5 x 10 <sup>3</sup>                 | 10 <sup>3</sup>                       | 4                |

| Weight (N) | Area (m <sup>2</sup> ) | Pressure (Nm <sup>-2</sup> ) |
|------------|------------------------|------------------------------|
| -          | 0.04                   | 20,000                       |
| 1500       | 500                    | -                            |

Answer:

Using the formula, pressure = weight/area:

| Weight (N) | Area (m <sup>2</sup> ) | Pressure (N.m <sup>-2</sup> ) |
|------------|------------------------|-------------------------------|
| 800        | 0.04                   | 20000                         |
| 1500       | 500                    | 3                             |

**6. The density of a metal is  $10.8 \times 10^3 \text{ kg/m}^3$ . Find the relative density of the metal.**

Solution:

Data: Density of the metal =  $10.8 \times 10^3 \text{ kg/m}^3$

density of water =  $10^3 \text{ kg/m}^3$

relative density of the metal = ?

Relative density of a substance

$$= \frac{\text{density of the substance}}{\text{density of water}}$$

$$= \frac{10.8 \times 10^3 \text{ kg/m}^3}{10^3 \text{ kg/m}^3} = 10.8$$

The relative density of the metal = 10.8.

**7. The volume of an object is  $20 \text{ cm}^3$  and the mass is 50 g. The density of water is  $1 \text{ gm}^{-3}$ . Will the object float on water or sink in water?**

Solution :

$$\rho (\text{object}) = \frac{50 \text{ g}}{20 \text{ cm}^3} = 2.5 \text{ g/cm}^3$$

It is greater than the density of water.

Hence, the object will sink in water.

**8. The volume of a plastic-covered sealed box is  $350 \text{ cm}^3$  and the box has a mass  $500 \text{ g}$ . Will the box float on water or sink in water? What will be the mass of water displaced by the box?**

Solution:

$$\rho (\text{box}) = \frac{500 \text{ g}}{350 \text{ cm}^3} = \frac{10}{7} \text{ g/cm}^3$$

It is greater than that of water.

Hence, the box will sink in water.

The volume of water displaced by the box ( $V$ ) = the volume of the box  
=  $350 \text{ cm}^3$

$$\text{Now, density } (\rho) = \frac{\text{mass } (m)}{\text{volume } (V)}$$

$$\therefore m = \rho V$$

$\therefore$  The mass of water displaced by the box =  $1 \text{ g.cm}^{-3} \times 350 \text{ cm}^3 = 350 \text{ g}$ .